Devashish Tupkary

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website

RESEARCH INTERESTS

Quantum Key Distribution, Quantum and Classical Cryptography, Open Quantum Systems.
5 peer-reviewed publications and 7 pre-prints.

ACADEMIC BACKGROUND

Ph.D. Physics (Quantum Information), GPA : 92/100 — 2023-Present
Institute for Quantum Computing, University of Waterloo
Mike and Ophelia Lazaridis Fellowship.

- Research on security analysis of practical quantum key distribution protocols.
- Relevant Coursework (Quantum Information and Cryptography): Quantum Error-correction and Fault-Tolerence, Quantum Information Processing, Implementation of Quantum Information Processing, Theory of Quantum Information, Applied Quantum Cryptography, Applied Cryptography, Mathematics of Public Key Cryptography, Privacy Data and Security.
- Contributor to Open QKD Security software package for numerical QKD analysis.

M.Sc. Physics (Quantum Information), GPA : 96/100 — 2020-2022
Institute for Quantum Computing, University of Waterloo

B.Sc. (Research) Physics, GPA : 9.2/10 — 2016-2020
Indian Institute of Science

- Graduated First Class with Distinction.
- Relevant Coursework (Quantum Information and Cryptography) : Quantum Mechanics 1, Quantum Mechanics 2, Quantum Computation, Quantum Measurements and Control, Quantum Information Theory, Information Theory, Theoretical Foundations of Cryptography.

PUBLICATIONS

[P1] ***Phase error estimation for passive detection setups with imperfections and memory effects***.
Zhiyao Wang, Devashish Tupkary, Shlok Nahar, arXiv link.

[P2] ***Imperfect detectors for adversarial tasks with applications to quantum key distribution***.

Shlok Nahar, Devashish Tupkary, Norbert Lütkenhaus, arXiv link. Under review at Quantum.

[P3] ***Security of quantum key distribution with source and detector imperfections through phase-error estimation***.
Guillermo Currús-Lorenzo, Margarida Pereira, Shlok Nahar, Devashish Tupkary, arXiv link.

[P4] ***QKD security proofs for decoy-state BB84: protocol variations, proof techniques, gaps and limitations***.
Devashish Tupkary, Ernest Y-Z Tan, Shlok Nahar, Lars Kamin, Norbert Lütkenhaus, arXiv link.

[P5] ***A consolidated and accessible security proof for finite-size decoy-state quantum key distribution***.
Jerome Wiesemann, Jan Krause, Devashish Tupkary, Davide Rusca, Nino Walenta, Norbert Lütkenhaus, arXiv link. Under review at Quantum.

[P6] ***Improved finite-size effects in QKD protocols with applications to decoy-state QKD***.
Lars Kamin, Devashish Tupkary, Norbert Lütkenhaus, arXiv link. Under review at Physical Review Research. Combined with other work, accepted talk at QCRYPT 2024 and shortlisted for best student theory submission.

[P7] ***Phase error rate estimation in QKD with imperfect detectors***.
Devashish Tupkary, Shlok Nahar, Pulkit Sinha, Norbert Lütkenhaus, arxiv link. Under review at Quantum. Combined with other work, accepted talk at QCRYPT 2024 and shortlisted for best student theory submission.

[P8] ***Postselection technique for optical prepare-and-measure QKD protocols***.
Shlok Nahar, Devashish Tupkary, Yuming Zhao, Norbert Lütkenhaus, Ernest Y.-Z. Tan, PRX Quantum 5, 040315. Combined with other work, accepted talk at QCRYPT 2024 and shortlisted for best student theory submission.

[P9] ***Security Proof for Variable-Length Quantum Key Distribution***.
Devashish Tupkary, Ernest Y.-Z. Tan, Norbert Lütkenhaus, Phys. Rev. Research 6, 023002. Combined with other work, accepted talk at QCRYPT 2024 and shortlisted for best student theory submission. Awarded the Cryptoworks21 Distinguished Student Paper Award.

[P10] ***Using Cascade in Quantum Key Distribution***.
Devashish Tupkary, Norbert Lütkenhaus, Phys. Rev. Applied 20, 064040.

[P11] ***Searching for Lindbladians obeying local conservation laws and showing thermalization***.
Devashish Tupkary, Abhishek Dhar, Manas Kulkarni, and Archak Purkayastha, Phys. Rev. A 107, 062216. Invited Talk at CQIQC Seminar, University of Toronto, 10th Feb 2023. video link.

[P12] ***Fundamental Limitations in Lindblad Descriptions of systems weakly coupled to baths***.

Devashish Tupkary, Archak Purkayastha, Abhishek Dhar, Manas Kulkarni. Phys. Rev. A 105, 032208.

SCHOLARSHIPS AND AWARDS

- Cryptoworks21 Distinguished Student Paper Award - For [PP9]. 2025
- Mike and Ophelia Lazaridis Fellowship - Fellowship during PhD, includes tuition and a yearly stipend of 36000 CAD. 2023-Present
- Long Term Visiting Student Program, International Centre for Theoretical Studies - Tata Institute of Fundamental Research - Among a dozen students selected for a year-long funded research project. 2019
- Madhava Mathematical Competition. *Rank 6* - A nationwide mathematics competition for undergraduate students. Attended the Madhava Mathematics Competition camp at the Indian Statistical Institute, Bangalore. 2017
- Kishore Vaigyanik Protsahan Yojana (KVPY-SA Fellowship). *Rank 39* - Fellowship during undergrduate degree, includes a yearly stipend of 80000 INR. 2016-2020
- National Bal Shree Honour for Creative Scientific Innovation - One of the three highest national honours for children by the Government of India. 2016
- IIT-JEE (Advanced). *Rank 312* - A nationwide entrance exam for college admissions to the Indian Institute of Technology (IITs). 2016
- Indian National Mathematics Olympiad - *Ranked among the top 70 nationwide* 2016
- Qualified Indian National Physics Olympiad - *Ranked among the top 35 nationwide. 2016*
- *National Talent Search Examination - Among 1000 students awarded the NTSE Scholarship nationally in each year. 2012-2016*

INVITED AND CONTRIBUTED TALKS

- Contributed Talk at TQC 2025 on "Phase error rate estimation in QKD with imperfect detectors." Based on [PP7]. 16 Sept 2025
- Invited Tutorial Talk at QCrypt 2025 on "Security Proofs for Decoy-State BB84 and Their Performance." Based on [PP4]. 26 August 2025
- Lecture at Cryptoworks21, University of Waterloo, on "An Introduction to Quantum Key Distribution." 22 July 2025
- Contributed Talk at QCrypt 2024 on "Security Proof for Variable-Length Quantum Key Distribution" and "Variable-Length QKD Security Proof for Imperfect Detectors through Phase-Error Estimation." Based on [PP9],[PP7]. 2 September 2024
- Invited Talk at CQIQC Seminar, University of Toronto, on "Lindbladians Obeying Local Conservation Laws and Showing Thermalization". Based on [PP11][PP12]. 10 February 2023

SEMINARS

- Talk at IISc Bengaluru, IISER Pune on "Extending Quantum Key Distribution security proofs to practical detectors". Based on [PP7][PP1][PP3].11th,12th Sept 2025
- Talk at IIT Bombay on "An Introduction to Quantum Key Distribution". 1st Sept 2025
- Talk at ICTS-TIFR on " A brief introduction to Quantum Key Distribution" 4th Jan 2024
- Contributed Talk at QKD Security Proofs workshop on "Phase error estimation with basis-efficiency mismatch." Based on [PP7] 31st Sept 2024
- Contributed Talk at QKD Security Proof workshop on "Postselection technique" and "Variable-length QKD." Based on [PP9][PP8] 12th,13th Sept 2023
- Contributed Talk at QKD Security Proof workshop on "Improved processing of classical data in QKD" 21st Sept 2022

SUPERVISION

- Supervised Zhiyao Wang, research project at IQC–Waterloo, leading to [PP1]. 2024–2025
- Supervised Jerome Wiesemann, research project leading to [PP5]. 2024–2025
- Supervised Soumyadeep Sarma, Bachelor's thesis at ICTS–TIFR, Bengaluru. 2024–Present

POSITIONS OF RESPONSIBILITY

- Co-Organizer of the QKD Security Proof Workshop at IQC Waterloo. 2022,2023,2024
- Co-Organizer of the weekly IQC Student Seminar Series. 2023-2024